last two speeches he has changed his phraseology to make clear that he means merely that there should be a definite statutory assurance that rates must be so fixed as to produce a prescribed rate of return. In his address before the Bankers' Association of New England he said:

"We ought to have a standard to determine what the railroad companies are entitled to. It ought to be fixed with such certainty that it will constitute an assurance to the railroad companies that they will be permitted to earn a sufficient amount to attract new capital. In order to do that it will be necessary, I believe, to have an official valuation of the railroads and a specified return upon that valuation Whatever return is earned in excess of the standard return should be divided between the public and the railroad, leaving enough for the railroad company to stimulate its initiative and its desire to economize, and yet giving the public a portion of it to get rid of this feeling that if there are undue profits they will inure exclusively to the private owners."

In reply to the argument that to complete the valuation and to work out his proposed consolidations of the railroads into a few systems will take too much time, Mr. Hines proposes an interim form of regulation of the railroads after their return to private management. On this point he says:

"I do not believe there was ever a better time in the history of the country than now to get a really constructive form of railroad regulation. And I think that the fact that it is going to take time to do it should not operate to defeat the purpose to do what the situation will require. It will take time; it can not be done this month or next month, it can not be done completely this year. There may have to be some interim form of regulation. The railroad systems ought to be turned back to the private management at the end of this year, as already indicated by the President, and yet an interim form of management and control could be devised which would carry forward the railroad operations until this thoroughly necessary and important constructive program can be carried out. But the machinery for carrying out the constructive program, the principles upon which it shall be carried out, ought, in my opinion, to be determined now while the public throughout the country has its mind centered on the problem and is trying to get a genuine and permanent solution."

Non-Lifting Injector Indicator

THE USUAL FORM OF tell-tale for non-lifting injectors warns the engineer by discharging a jet of steam into the cab when the injector blows back. An indicator has been developed by William Sellers & Company, Inc., Philadelphia, Pa., which not only performs the function of the tell-tale without discharge of steam into the cab, but in addition advises the engineer if the injector is wasting water at the overflow.

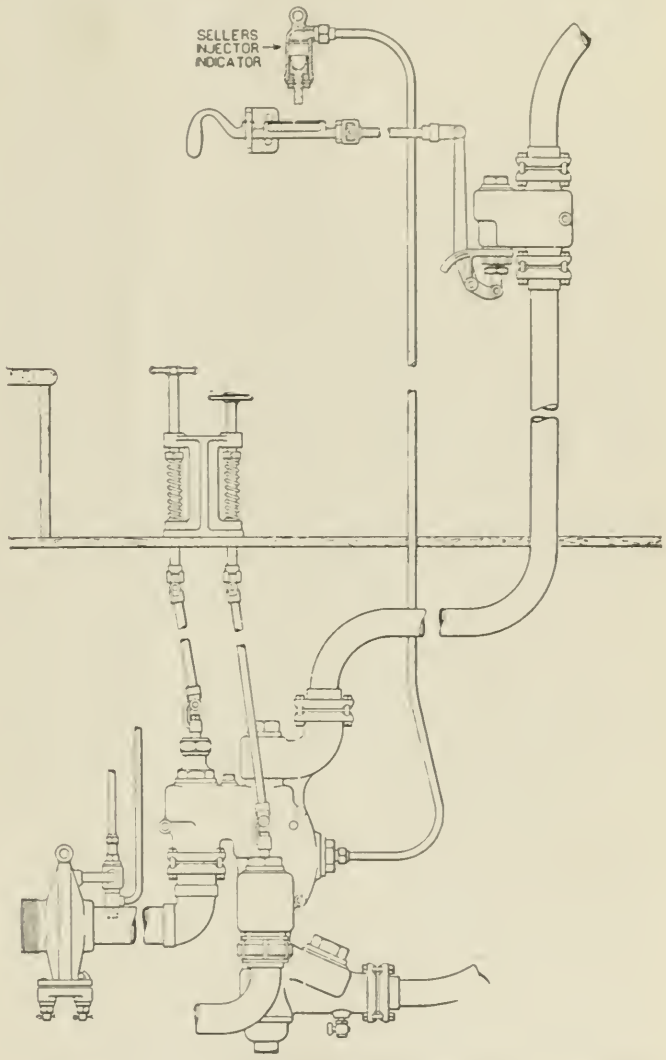
The device consists of a vertical cylinder 4 in. long, containing a loosely fitting piston, seating at each end of its stroke and provided with a projecting end, visible to the engineer when resting on its lower seat. The upper end of the cylinder is connected by a ½-in. copper pipe with the overflow chamber of the injector.

Its action depends upon the well-known principle of the working of the injector, that when feeding without waste there is always a partial vacuum within the overflow chamber of an injector of the gravity overflow type. This vacuum is utilized to raise the loosely fitting piston to its upper seat, causing the projecting plug to disappear from view.

If for any cause the injector starts to waste, the vacuum within the overflow chamber is broken, the piston drops and seats at the lower end of its stroke, exposing the projecting plug. The operator is thus warned and will partially close the lazy cock, stopping the waste. If there is interruption of the water or steam supply, and the injector "flies off," the piston will instantly drop to its lower seat, preventing outflow of steam and warning the engineer of the danger.

When the injector operating valve is open, the end of the piston must always be invisible, unless heating the water in the tank. If steam is blown back into the tank in winter, the position of the projecting plug of the indicator is a continuous reminder that the injector is not operating and there may be danger of overheating the water.

A further advantage of this form of indicator is that it enables the engineer or fireman to obtain the actual minimum capacity. Most operators do not regulate the water supply closely, fearing that the injector may break off without warning or waste at the overflow. The indicator is so sensitive to conditions in the overflow chamber of the injector that the exact minimum can be obtained by regulating the lazy cock until the end of the piston appears, due to the loss of vacuum in the overflow chamber. It is claimed that a very slight opening of the lazy cock will cause the piston to rise and the exact minimum to be obtained.



Application of the Sellers Non-Lifting Injector Indicator

The application of the indicator also will reduce the tendency of the engineer to close the heater valve, which prevents the injector from restarting automatically. It is designed to make the non-lifting injector as safe and certain to operate as an open-overflow lifting injector.

The indicator is applicable to the Sellers, Nathan WF, Nathan Simplex, Chicago and other forms using an overflow chamber which contains the combining tube and is closed against the atmosphere by a gravity overflow valve.